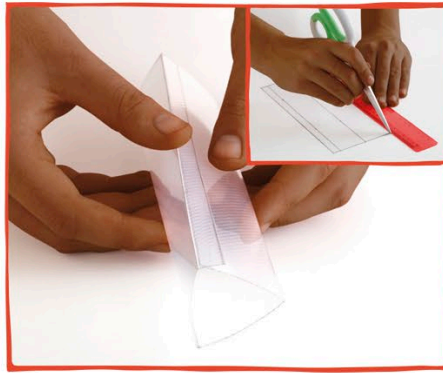




3 On the plastic folder, draw a rectangle as long as the tube and two and a half times as wide as the tube's diameter. Draw lines to divide the rectangle into three equal parts. Draw a narrow tab on one side.



4 Cut out the rectangle, then score along the three inside lines with the scissors, using the edge of the ruler. Fold along the scored lines to make a triangular prism. Stick down the tab with tape.



5 Insert your prism into the cardboard tube, so that it rests against the paper circle at the end. It should fit snugly, but if not, use small pieces of tape to secure it in place.



6 Place plastic wrap loosely over the open end of the tube, and stick it in place with tape. Now put some colored beads on top of the plastic wrap.



7 Cut a circle of wax paper, wider than the tube. Place it over the beads, then cut slits in the edge of the paper and tape it to the tube.



8 Decorate the tube, if you like. Now look through the hole in the paper circle. Point the tube toward a window or light, and turn it around. Enjoy the show!

HOW IT WORKS

At the center of the pattern you produced by your kaleidoscope is the bead-filled triangle at the end of the cardboard tube. You can see it directly because some of the light that passes through the beads travels straight through the tube. Reflections around the central triangle are made by light that has reflected off one or more of the three shiny surfaces inside the prism. Each surface acts as a mirror, changing the direction the light travels, and so making it appear to have come from behind the mirror.

